



A project report on

MINI DRILLING AND CUTTING MACHINE

submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology
in
Mechanical Engineering

by

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CERTIFICATE

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The Project was evaluated by us on

EXAMINER 1
EXAMINER 3

EXAMINER 2
EXAMINER 4

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ABSTRACT

The mini drilling and cutting machine was designed to meet the needs of small workshops, educational labs, and DIY enthusiasts who need a compact, affordable, and easy-to-use tool. This project brings together two essential functions—drilling and cutting—into one portable unit, making it easier to handle basic fabrication tasks without the need for separate machines. Built with a lightweight PVC frame, a reliable DC motor, and an adjustable chuck, the machine is simple to operate and built to last. Power is controlled through a PWM controller and a battery management system (BMS), ensuring both safety and flexibility during use. During testing on materials like plastic and wood, the machine delivered smooth, precise cuts with minimal vibration. It performed consistently across multiple trials, showing low power consumption and dependable results. This compact dual-purpose tool is a practical solution for anyone working with limited space or budget—perfect for quick prototyping and small fabrication jobs.

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LIST OF SYMBOLS / ABBREVIATIONS

Symbol / abbreviations	Description
DC	Direct Current
PVC	Polyvinyl Chloride
BMS	Battery Management System
PMW	Pulse Width Modulation
V	Voltage
A	Ampere

LIST OF PHOTOGRAPHS

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1. CHAPTER I: INTRODUCTION

1.1 Overview

In today's fast-paced world of prototyping and small-scale manufacturing, there's a growing need for compact, versatile tools that can handle multiple tasks with ease. One common requirement is a machine that can both drill and cut soft materials accurately, without taking up too much space or breaking the bank. That's exactly what the mini drilling and cutting machine is built for—it's a simple, portable, and affordable solution. Whether you're working in a DIY setup, an educational lab, or a small fabrication unit, this tool is designed to make your work easier and more efficient.

1.1.1 Need for the Project

Traditional workshop tools are often large, power-hungry, and require separate setups for drilling and cutting. This not only takes up more space but also adds to the cost and time involved. By combining both functions into a single, compact machine, the process becomes much more efficient. It's especially useful in educational settings, where hands-on learning is important but space and resources are often limited.

1.1.2 Challenges in the Domain

- Lack of affordable dual-function machines.
- Limited availability of compact designs for multi-purpose tasks.
- Safety and stability concerns in low-cost equipment.

1.2 Objectives

Aim:

To design and develop a compact and versatile mini drilling and cutting machine for efficient material processing.

Objectives:

- To fabricate a machine that integrates both drilling and cutting mechanisms.
- To ensure the machine is low-cost, user-friendly, and safe to operate.
- To test the machine on different materials and evaluate its performance.

2. CHAPTER II: BACKGROUND

2.1 Literature Review & Design Gap

- Most traditional workshop tools were designed to handle just one task at a time. While there are some machines that try to combine multiple functions, they often end up sacrificing either power or precision. For DIY projects and academic prototyping, there's still a clear gap when it comes to affordable, multi-purpose tools that can do the job well without cutting corners.

2.2 Theoretical Background

The primary principles used in this design include:

- Rotational mechanics via DC motors.
- Chuck-based clamping for drilling accuracy.
- Basic electronics for speed control (PWM) and power safety (BMS).
- Material selection based on mechanical properties and cost-efficiency.

3. CHAPTER III: IMPLEMENTATION

3.1 Design Approach

Traditional workshop tools are usually built to do one job at a time. While some machines try to combine different functions, they often fall short—either they lack enough power or they don't deliver the level of accuracy needed. For students, hobbyists, and makers working on DIY or academic projects, there's still a real need for low-cost, all-in-one tools that are both reliable and efficient.

3.2 Materials and components Used

- **PVC Pipes** – Selected for their lightweight, corrosion resistance, and ease of fabrication.



- **DC Motor** – Acts as the core driver for both cutting and drilling actions.



- **Chuck** – Used to securely mount drill bits.



- **PWM DC Motor Speed Regulator** – Allows control of motor speed depending on the material type.



- **Battery Management System (BMS)** – Ensures safe operation and efficient use of battery power.



- **Drill bits and cutters:** Drill bits are used for precise drilling operations in various materials, and Cutters for cutting through different materials, such as plastic, wood, and foam.



- **DC Jack Power Connector** – Facilitates easy connection to external power sources.



- **Batteries** – Provide the necessary power for the operation of the machine, ensuring portability.



- **Switch** – Used for turning the machine on and off for safe operation.



- **Power Adapter** – Supplies a stable and regulated power source when operating without batteries.



3.3 Assembly and Testing

All the components were carefully assembled onto the PVC frame, making sure everything was properly aligned for precise drilling and cutting. Safety was a key focus—wiring was well insulated, and the motor was securely mounted to keep the setup stable during operation.

3.4 Impact Assessment

Social Factors

- Stakeholders include students, hobbyists, and small workshop owners.
- The machine aligns with the need for affordable and safe fabrication tools.
- It promotes skill development in academic institutions and self-learning setups.

Environmental & Sustainability Factors

- No hazardous emissions; minimal noise and dust generation.
- Lightweight PVC reduces the environmental load compared to metal-based machines.
- Power-efficient operation via DC components and battery management.

Budget allocation

Expenditure head	Estimated Expense	Remarks
Travel	Rs 300	
Study Resources		
Materials for experiments	Rs 3270	
Report preparation	Rs 30	

Any other		
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CHAPTER IV: RESULTS AND DISCUSSION

4.1 Performance Analysis

The machine was tested on various soft materials including:

- Plastic sheets
- Foam boards
- Thin plywood

4.2 Observations

Performance of the Mini Drilling and Cutting Machine on Various Materials:

Material	Drilling Accuracy	Cutting Quality	Remarks
Plastic	High	Smooth edges	Stable
Foam	High	Clean cut	Fast
Plywood	Moderate	Minor vibration	Manageable

- Demonstrated **low vibration** and **minimal noise** during operation.
- **Cutting edges** were consistently smooth, and **drilling depths** were precise.
- The machine remained stable and functional over **multiple testing cycles**.

4.3 Key Highlights

- Successfully combined cutting and drilling in one unit.
- Power usage remained low, ideal for battery or small DC supply.
- Easy to operate, even for untrained users.
- No mechanical or electrical failures observed during extended trials.

5. CHAPTER V: CONCLUSION & SCOPE FOR FURTHER WORK

5.1 Conclusion

The mini drilling and cutting machine successfully met its main goal of being a compact, dual-purpose, and budget-friendly tool for basic fabrication. It performed reliably on different soft materials and was easy to operate, even for beginners. With its simple design and consistent results, this device is a great fit for small workshops and educational labs where space and resources are limited.

5.2 Scope for Future Work

- **Automation:** Integration of microcontrollers for auto-drill or timed cutting actions.
- **Brushless Motors:** Adoption of BLDC motors to reduce wear and extend lifespan.
- **Material Capability:** Modifications to work on tougher materials like acrylic or thin metal sheets.
- **Safety Enhancements:** Adding enclosures and a dust extraction system for safer use.

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